SIMATS ENGINEERING

ASSESSMENT TOOL 2: Application-Based Questions

COURSE NAME: **Embedded Systems**
COURSE CODE: **ECA14**

COURSE OUTCOME COVERED

CO 3: ***Analyze RTOS-based task scheduling, synchronization, and communication mechanisms to achieve reliable embedded system performance.(BT5)***

Assessment Tool : Application-Based Questions

Assessment Tool Weightage : 30%
Assessment Tool : Application-Based Questions
Duration : 30 minutes
Marks : 45
Type of Assessment conduction : Hybrid
Topics Covered : Principles of RTOS, Semaphores and Queues, Task and Task States, Tasks and Data, Semaphores, and Shared Data; Message Queues

Description

Each student has to answer three questions.

Students will independently answer one application-based question, demonstrating their ability to apply concepts, analyze practical scenarios, justify solutions, communicate technical ideas effectively, and relate theoretical knowledge to real-world embedded system applications.

Assessment will be based on rubric-based evaluation.

Evaluation Type : Application-Based Assessment

Result Generation : Rubric-based assessment

QUESTIONS: 25 **Total Marks: 15**

S. No.	Question Bank	BTL	Marks
1	A smart traffic signal system must give priority to emergency vehicles. Illustrate how tasks, events, and priorities can be applied to achieve this behavior.	BL3	15
2	In an industrial system, multiple sensors update a shared memory location. Demonstrate how semaphores can be applied to avoid data inconsistency.	BL3	15
3	A wearable health monitor samples heart rate periodically and triggers alerts when abnormal. Show how timers and events can be used to implement this.	BL3	15
4	A robotic system has separate sensing and control tasks. Illustrate how message queues can be used for communication between these tasks.	BL3	15
5	In a home automation system, multiple tasks access shared configuration data. Apply appropriate synchronization techniques to ensure safe access.	BL3	15
6	A real-time system requires one task to wait for data from another before execution. Apply a suitable IPC mechanism and justify its use.	BL3	15
7	A temperature monitoring system samples data every 2 seconds. Demonstrate how timer functions can be used for this requirement.	BL3	15
8	A system shows inconsistent output due to concurrent data access. Apply synchronization techniques to resolve the issue.	BL3	15
9	In an automotive control system, high-priority tasks must execute without delay. Show how task states and scheduling enable this.	BL3	15
10	A communication system transfers data between tasks continuously. Apply message queues to ensure reliable communication.	BL3	15
11	Multiple tasks must respond to a single external trigger. Demonstrate how event mechanisms can be used for synchronization.	BL3	15
12	A system contains both periodic and event-driven tasks.	BL3	15

	Apply appropriate RTOS mechanisms to manage their execution.		
13	In a shared data environment, tasks frequently read and write common variables. Apply mutual exclusion techniques to maintain data integrity.	BL3	15
14	A drone system processes sensor inputs and responds in real time. Show how task prioritization and IPC improve system performance.	BL3	15
15	A system encounters deadlock due to improper semaphore usage. Illustrate how the issue can be identified and resolved.	BL3	15
16	A smart irrigation system activates watering based on soil conditions. Apply event-driven execution to implement this functionality.	BL3	15
17	A system requires fast communication between tasks with minimal overhead. Choose between mailbox and queue, and justify your selection.	BL3	15
18	A critical task must meet strict deadlines in a multitasking system. Apply RTOS scheduling principles to ensure timely execution.	BL3	15
19	A logging system collects data from multiple tasks. Demonstrate how synchronization and communication mechanisms can be applied.	BL3	15
20	Two dependent tasks must execute in sequence. Apply appropriate synchronization techniques to enforce this order.	BL3	15
21	A battery-powered embedded system must reduce power consumption. Show how task scheduling and event handling contribute to this goal.	BL3	15
22	A safety system must immediately trigger an alarm when a fault is detected. Apply events and task prioritization to achieve this.	BL3	15
23	A system needs to perform periodic health checks along with real-time processing. Demonstrate how timers can be integrated.	BL3	15
24	A communication system experiences delays due to inefficient IPC. Apply optimization techniques using	BL3	15

	RTOS mechanisms.		
25	A real-time system handles multiple concurrent operations. Show how tasks, semaphores, and IPC mechanisms can be integrated effectively.	BL3	15

Evaluation Rubric (15 Marks per Question – Application-Based Question

Criteria	Weightage	Exemplary	Proficient	Developing	Beginning
Understanding of the Problem Statement	20%	3	2	1	0
Application of RTOS Concepts	30%	5	4	2–3	0–1
Implementation / Problem Solving Approach	20%	4	3	1–2	0
Accuracy of Solution	20%	3	2	1	0
Clarity in Presentation	10%	3	2	1	0
Total Marks	**100%**	**18**	**13**	**6–8**	**0–1**

Performance Level Description

Level	Description
Exemplary	Complete understanding and accurate application of RTOS concepts with clear implementation and justification
Proficient	Good understanding with minor mistakes in application or explanation
Developing	Partial understanding with incomplete implementation or limited clarity
Beginning	Minimal understanding with incorrect or missing RTOS concepts

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: